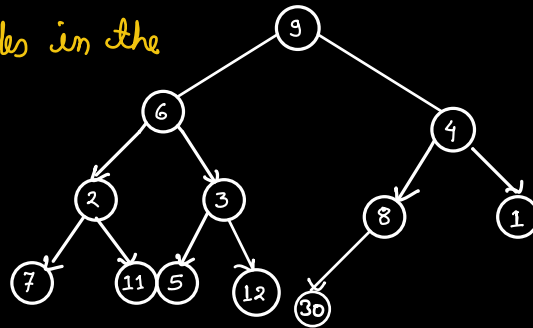


HW Q Given a BT where all levels are completely filled except possibly the last level.

Google In the last level all nodes are left aligned.

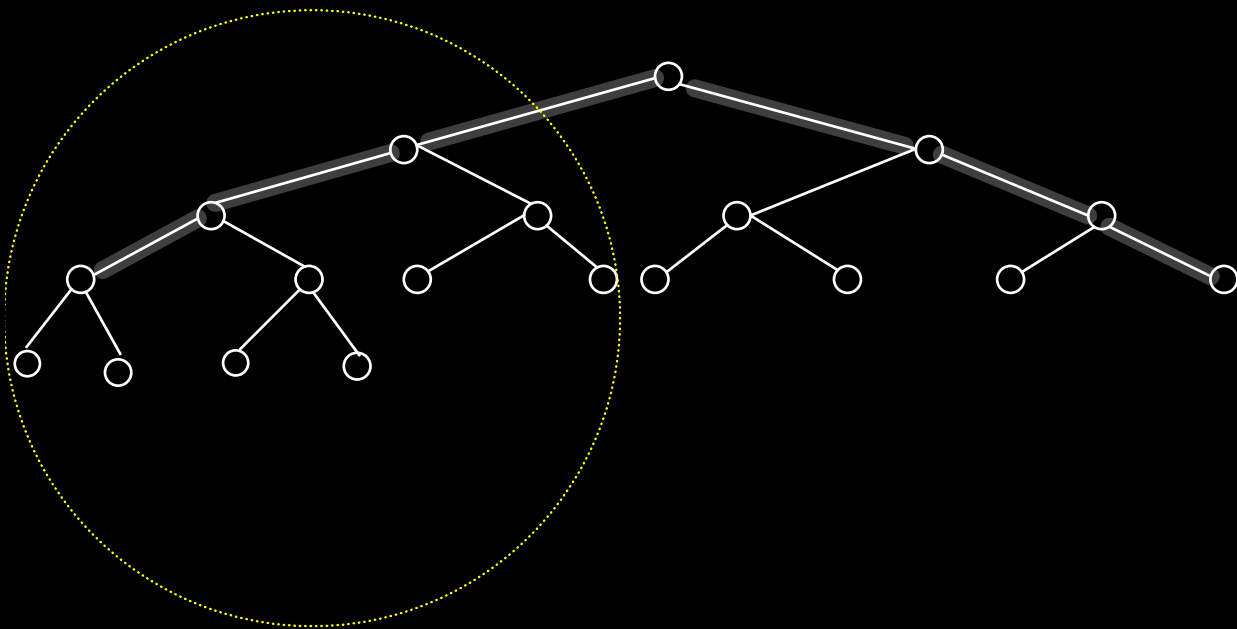
(Complete Binary Tree)

Count the number of nodes in the given BT.

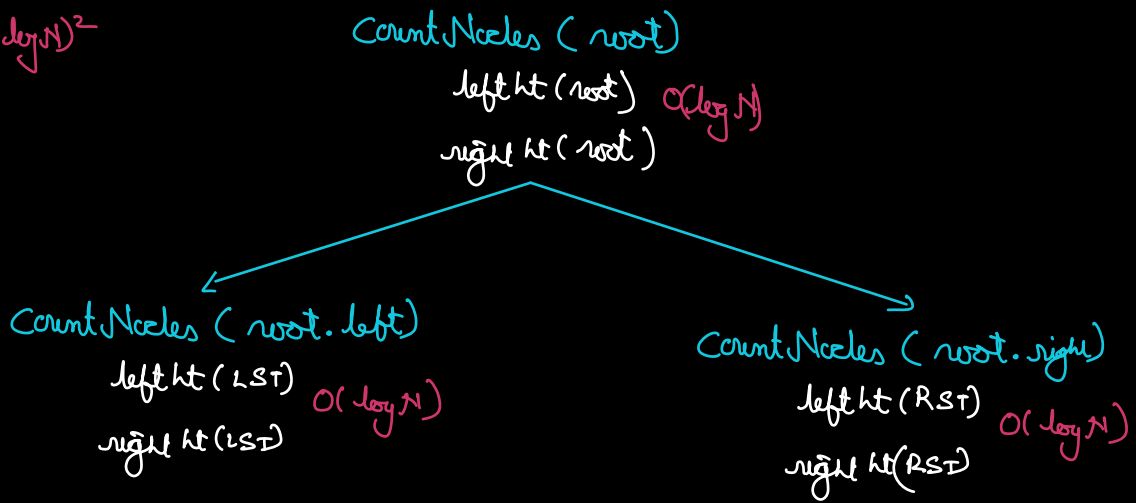


→ 12

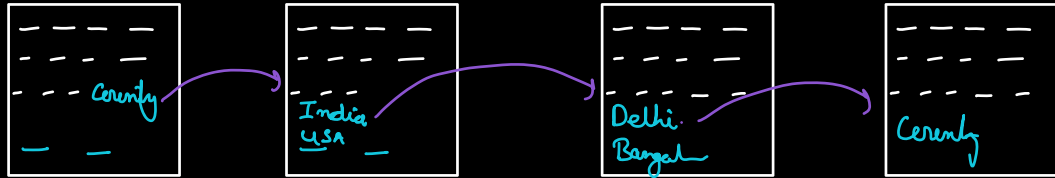
No of nodes in a completely filled tree = $2^{(h+1)} - 1$
(even the last level filled)



TC: $O(\log N)^2$



Google



HashSet

SC : $O(N \cdot l)$

No of links → length of link

ib.com / practice

ib.com / dsa

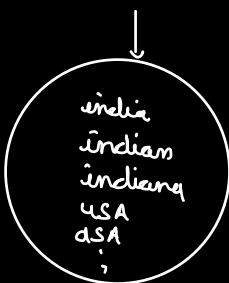
ib.com / stack

gfg.com / stack

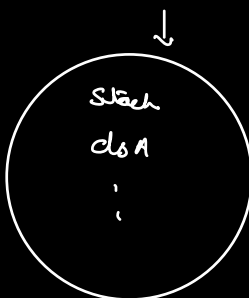
gfg.com / dsa

HashMap < String, HashSet < String > >

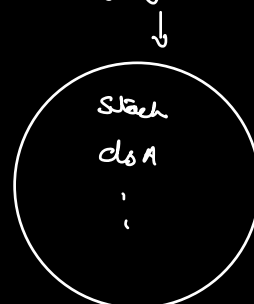
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 interviews Kick Start . com
 interviews Boss . com
 interviews Ready . com

a
 abc . com
 abcd . com
 ade . com
 aab . com
 aap . com

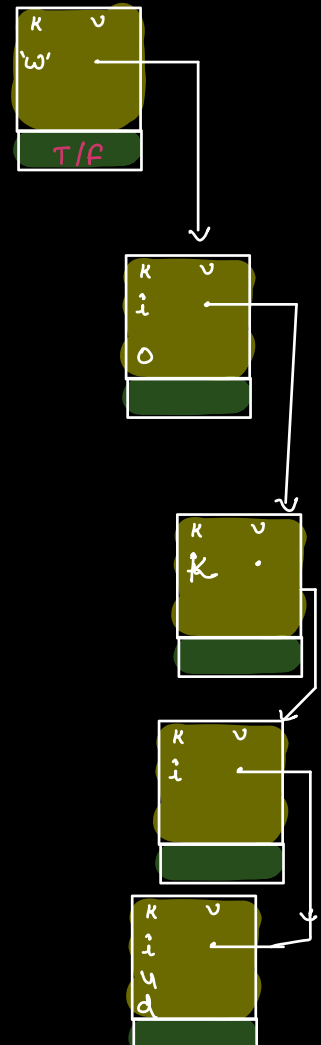
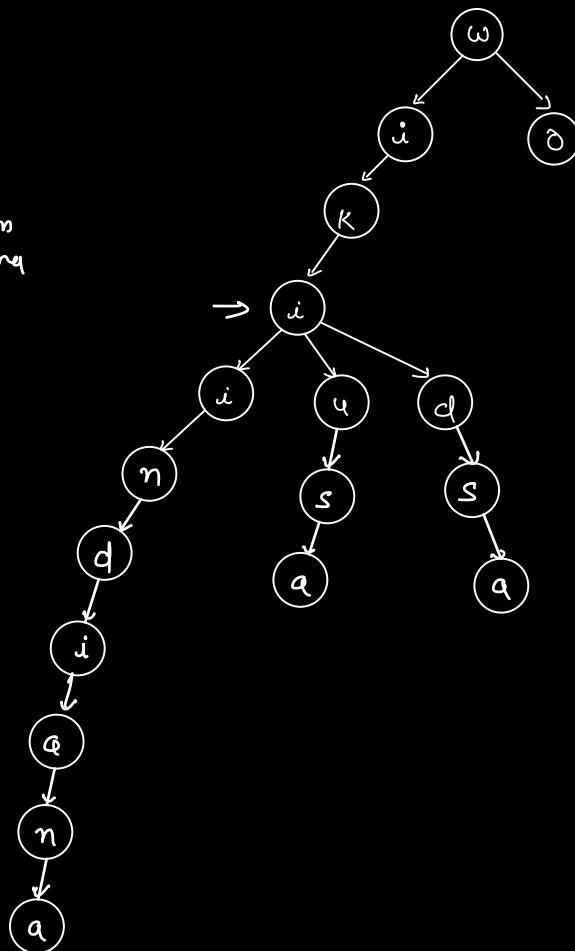
b

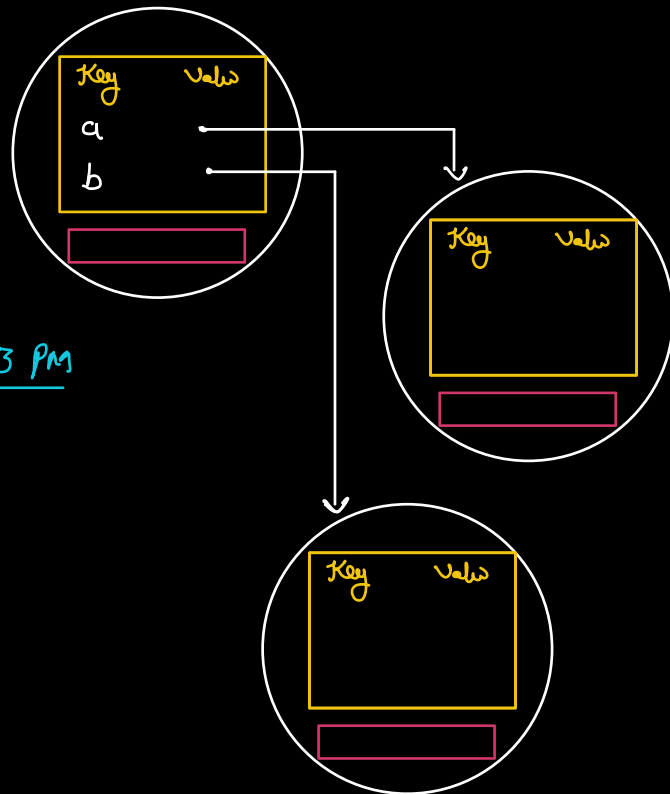
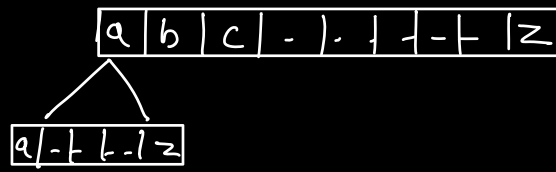
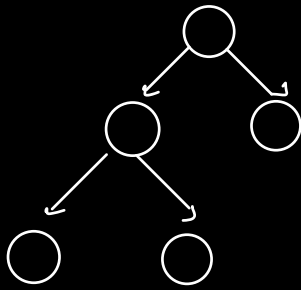
c

```

class TrieNode {
    HashMap <Char, Node> map;
    boolean isEnd;
}
  
```

india
 indian
 indian
 USA
 USA
 world → map
 maps
 cat
 cats
 dog
 doggo





Break till 11:03 pm



RANKER ✓
RANDOM ✓
MESSI ✗
RAND ✗
FIREWALL ✗
FIR ✗
FRAND ✗
FIRES ✓
RAN ✗ ✓

RANKER ✓
RANDOM ✓
MESSI ✗
RAND ✗
FIREWALL ✗
FIR ✗
FRAND ✗
FIRES ✓
RAN (✗) ✓

```

→ cur = root;
for (i=0; i < word.length(); i++) {
    ch = word.charAt(i);
    if (! cur.map.containsKey(ch)) {
        cur.map.put(ch, new TrieNode());
    }
    cur = cur.map.get(ch);
}
cur.isEnd = true;

```

$T.C: O(L)$

Tc: $O(\underline{l})$

Search

→ `cur = root;`

`for (i=0; i < word.length(); i++) {`

`ch = word.charAt(i);`

`if (! cur.map.containsKey(ch)) {`

`return false;`

`}`

`cur = cur.map.get(ch);`

`}`

`return cur.isEnd;`

TC: $O(L)$

Prefix Tree

Trie